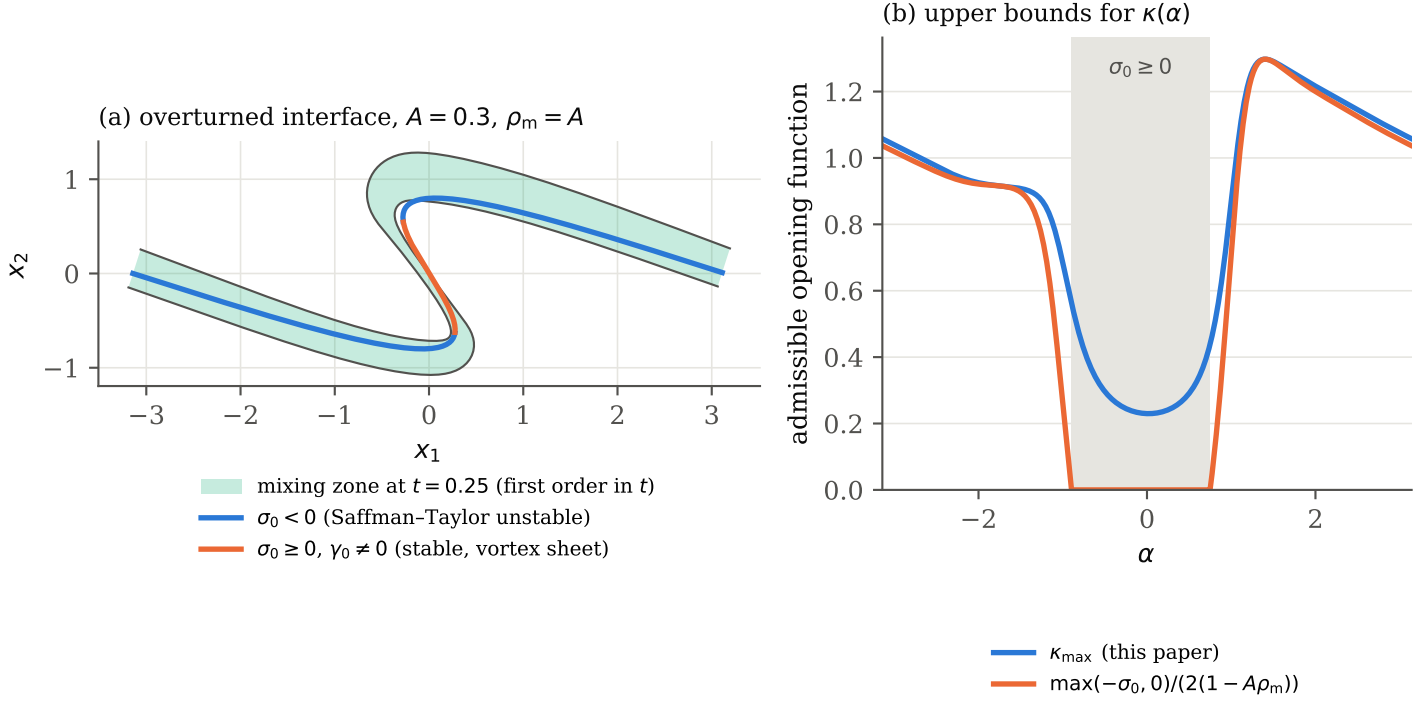
(a) overturned interface, $A = 0.3$, $\rho_m = A$
x_2
x_1
mixing zone at $t = 0.25$ (first order in t)
$\sigma_0 < 0$ (Saffman–Taylor unstable)
$\sigma_0 \geq 0$, $\gamma_0 \neq 0$ (stable, vortex sheet)
(b) upper bounds for $\kappa(\alpha)$
$\sigma_0 \geq 0$
admissible opening function
α
κ_{max} (this paper)
$\max(-\sigma_0, 0)/(2(1 - A\rho_m))$